

Experiment No. 18

Write a Prolog Program for STUDENT-TEACHER-SUBCODE

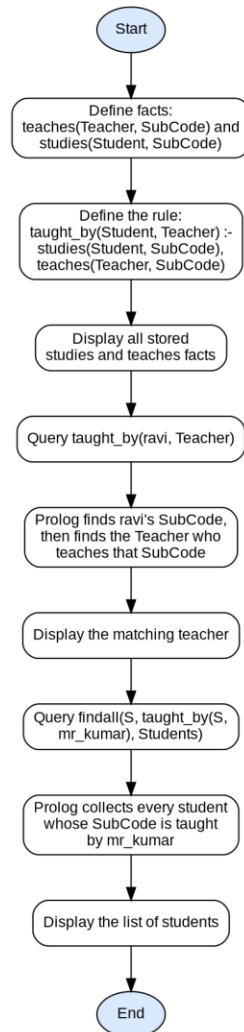
Aim

To write a Prolog program that stores facts about which teacher teaches which subject code and which student studies which subject code, and derives who teaches a given student using a rule that joins the two relations.

Algorithm

1. Start.
2. Define facts teaches(Teacher, SubCode) for each teacher-subject pairing.
3. Define facts studies(Student, SubCode) for each student-subject pairing.
4. Define the rule taught_by(Student, Teacher) :- studies(Student, SubCode), teaches(Teacher, SubCode), which joins the two relations through the shared subject code.
5. Display all the stored studies and teaches facts.
6. Query taught_by(ravi, Teacher) to find who teaches a given student.
7. Prolog finds ravi's subject code, then finds the teacher who teaches that subject code.
8. Display the matching teacher.
9. Query findall(S, taught_by(S, mr_kumar), Students) to find every student taught by a given teacher.
10. Prolog collects every student whose subject code is taught by mr_kumar into a list.
11. Display the list of students.
12. Stop.

Flowchart



Program

```

teaches(mr_kumar, cs301).
teaches(mrs_priya, cs302).
teaches(mr_arjun, cs303).

```

```

studies(ravi, cs301).
studies(anitha, cs302).
studies(suresh, cs301).
studies(kavya, cs303).

```

```

taught_by(Student, Teacher) :-
    studies(Student, SubCode),
    teaches(Teacher, SubCode).

```

main :-

```
format("STUDENT - TEACHER - SUBCODE Database~n"),
format("-----~n"),
forall(studies(S, Sub), format("~w studies ~w~n", [S, Sub])),
format("~n"),
forall(teaches(T, Sub), format("~w teaches ~w~n", [T, Sub])),
format("~nQuery: taught_by(ravi, Teacher)~n"),
taught_by(ravi, Teacher1),
format("Result: ravi is taught by ~w~n", [Teacher1]),
format("~nQuery: find all students taught by mr_kumar~n"),
findall(S2, taught_by(S2, mr_kumar), Students),
format("Result: ~w~n", [Students]).
```

Result

The program was executed successfully and the output was verified as correct.

?- main.

STUDENT - TEACHER - SUBCODE Database

ravi studies cs301
anitha studies cs302
suresh studies cs301
kavya studies cs303

mr_kumar teaches cs301
mrs_priya teaches cs302
mr_arjun teaches cs303

Query: taught_by(ravi, Teacher)

Result: ravi is taught by mr_kumar

Query: find all students taught by mr_kumar

Result: [ravi,suresh]

true.

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal:

The screenshot shows the Visual Studio Code interface with a file explorer on the left and a main editor area. The file explorer shows a project named 'AI-Lab' with files: 'sum_n.pl', 'name_dob.pl', 'student_teacher.pl' (selected), 'planets.pl', and 'hanoi.pl'. The main editor displays the code for 'student_teacher.pl'.

```

1 teaches(mr_kumar, cs301).
2 teaches(mrs_priya, cs302).
3 teaches(mr_arjun, cs303).
4
5 studies(ravi, cs301).
6 studies(anitha, cs302).
7 studies(suresh, cs301).
8 studies(kavya, cs303).
9
10 taught_by(Student, Teacher) :-
11     studies(Student, SubCode),
12     teaches(Teacher, SubCode).
13
14 main :-
15     format("STUDENT - TEACHER - SUBCODE Database~n"),
16     format("-----~n"),
17     forall(studies(S, Sub), format("~w studies ~w~n", [S, Sub])),
18     format("~n"),
19     forall(teaches(T, Sub), format("~w teaches ~w~n", [T, Sub])),
20     format("~nQuery: taught_by(ravi, Teacher)~n"),
21     taught_by(ravi, Teacher1),
22     format("Result: ravi is taught by ~w~n", [Teacher1]),
23     format("~nQuery: find all students taught by mr_kumar~n"),
24     findall(S2, taught_by(S2, mr_kumar), Students),
25     ! (code continues — see Program section)

```

The bottom panel shows the 'TERMINAL' output:

```

PS C:\Users\Athavan\AI-Lab> swipl student_teacher.pl
Welcome to SWI-Prolog (threaded, 64 bits, version 9.0.4)

?- main.
STUDENT - TEACHER - SUBCODE Database
-----
ravi studies cs301
anitha studies cs302
suresh studies cs301
kavya studies cs303
mr_kumar teaches cs301
mrs_priya teaches cs302
mr_arjun teaches cs303
Query: taught_by(ravi, Teacher)
Result: ravi is taught by mr_kumar
Query: find all students taught by mr_kumar
Result: [ravi,suresh]
true.
?-

```

The status bar at the bottom indicates the current file is 'main', line 1, column 1, using the Prolog dialect, UTF-8 encoding, LF line endings, and 4 spaces for indentation.